Project 1

Description:
The aim of the project is to analyze a logistic regression model in a situation where the training dataset contains observations with missing labels. We consider a binary classification problem in which Y is the response variable (taking values 0 or 1), and X is the feature vector.

The mechanism generating missing data in the response variable is described by the variable S:

- A value of S = 1 indicates that missing data has occurred; in this case, X is observed, while Y is not observed. We adopt the convention that missing data is denoted as Y = −1.
- A value of S = 0 indicates that no missing data has occurred; in this case, the variable Y (with value 0 or 1) is observed for that observation.

Example data are presented in the table below (the grey part serves as the training data).

X1	**X2**	**Y_{obs} (observed)**	**Y (true, unobserved)**	**S**	**Missing Y**
3	4	1	1	0	no
1	0.5	0	0	0	no
0.75	1	1	1	0	no
2	6	1	1	0	no
1	0	-1	1	1	yes
7	2.5	-1	0	1	yes

Different missing data mechanisms are considered:

1. **MCAR (Missing Completely at Random, MCAR):** P(S=1|X,Y)=P(S=1)=c. The labeling mechanism is completely random and does not depend on either X or Y.
2. **MAR (Missing at Random, MAR):** P(S=1|X,Y)=P(S=1|X) .The labeling mechanism depends only on X.
3. **MNAR (Missing Not at Random):** P(S=1|X,Y).The labeling mechanism may depend on both X and Y.

Task 1

The aim of the first task is to collect and prepare data sets for conducting experiments.

- Find **4 different real datasets** corresponding to classification problem with binary class variable containing numerical variables. Datasets should contain a relatively large number of numerical features to ensure sufficient model complexity.
    - You can use repositories: https://archive.ics.uci.edu/, https://www.openml.org/ or other sources.
    - Non-standard, interesting datasets will be appreciated. You can convert multi-class datasets to binary datasets by combining classes (for example, treat the majority class as positive and assign the remaining observations to the negative class).
    - Prepare datasets to run logistic regression:
        - filling in missing values
        - removing collinear variables,
- Propose and implement four Y-missing data generation schemes based on a fully labeled dataset. The input should consist of data for which we observe (X, Y) for every object. The function should return data of the form (X, Y_{obs}).

  Schemes:

    - MCAR should correspond to the MCAR setting.
    - MAR1 should correspond to the MAR setting and assume that the missing data mechanism depends only on a single explanatory variable.
    - MAR2 should correspond to the MAR setting and assume that the missing data mechanism depends on all explanatory variables.
    - MNAR should correspond to the MNAR setting.

Task 2

The aim of this task is own Implementation of logistic regression using optimization algorithm:

- **FISTA** (Fast Iterative Shrinkage-Thresholding Algorithm); solving Logistic Lasso Regression

These algorithms should work with data without missing values in X and Y.

- The input of the method: training data (features X_train and labels y_train), validation data (features X_valid and labels y_valid).
- For both algorithms, the optimal value of the lambda parameter should be selected by optimizing the appropriate measure on the validation set, separately for each algorithms. The user should be able to choose the following measures: recall, precision, F-measure, balanced accuracy (for a threshold of 0.5), area under the ROC curve, area under the sensitivity-precision curve.

- The following methods should be implemented:
    - fit(X_train,y_train),
    - validate(X_valid, y_valid, measure),
    - predict_proba(X_test),
    - Plotting the evaluation:
        - plot(measure, ...), which produces plot showing how the given evaluation measure changes with lambda,
        - plot_coefficients(...), which produces plot showing the coefficient values as function of lambda parameter.
- The part of the solution is discussing own implementation with sklearn imple
- mentation. For FISTA use penalty='l1'

  https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html

Task 3

Propose and implement a class **UnlabeledLogReg** that uses the data (X, Y_{obs}), i.e., both labeled observations and observations with a missing response value. The **UnlabeledLogReg** method should be based on own implementations of logistic regression (FISTA), although the detection of missing labels may be performed using any suitable techniques.

In this task, your main goal is to **propose two different algorithms** for constructing the complete vector of Y and then run logistic regression for fully observed data.

- In implementation of **UnlabeledLogReg,** user should select (at least, you can include also here Naive method, described below) method used to train logistic regression: FISTA.
- In **UnlabeledLogReg,** user should be able to select one of two implemented algorithms of completing Y
- In experiments compare three approaches:
    - The **UnlabeledLogReg** approaches
    - The **Naive** method: logistic regression based only on labeled data, i.e., observations for which S = 0.
    - The **Oracle** method: logistic regression based on (X, Y). This method serves as a reference benchmark and assumes that labels are available for all observations.
- As evaluation metrics, the following should be used: accuracy, balanced accuracy, F1-score, and ROC AUC. All these metrics should be computed on the test set. Important: in our experiments, missing data affects only the training set.
- Compare the three methods under four missing data generation schemes: MCAR, MAR1, MAR2, MNAR.
- Analyze the performance of the three methods for different values of the parameter

c in the MCAR setting.

Final grade

The final grade of project 1 you can get 50 points, and they are divided into three components:

- Report – **40 points**.
- Code – **5 points**
- Presentation in the form of an interview with the lecturer at the project class – **5 points**

Requirements for reports

Every point described below should be included in the separate section. **Maximal length of report is 6 pages A4 (title page, table of contest, appendix and references are not included in the limit).**
Report should include:

- Discussion of correctness of implementation of FISTA algorithm
- Methodology used in **UnlabeledLogReg** implementation
- Analysis of experiments described in Task 3. One part of the assessment is the scope of the experiments.

Requirements for code

- Instructions to run algorithm (README)
- Script, notebook to run the algorithm for new data
- Documentation of the functions - docstrings
- Readability of code and use of good practices for granularity of functions

Additional remarks:

- The projects are implemented in teams of 3 students. Teams can be formed between project groups, but each team will be assigned to one class term and instructor
- The projects are implemented in Python or R.

Organization of work

- Project consultations are held every two weeks. Since project classes are not mandatory, the team reports attendance at consultations in the Consultations form, and in addition, we ask for a legal message to the corresponding instructor. Consultation attendance should be reported by the preceding Tuesday by 6 pm.

- The teams' composition should be entered into the Teams file. The preferred instructor should also be completed. Teams can be completed by 01.03.2026. After this date we will make available the final assignment of teams to instructors and class dates.

- Schedule and topics for Project 1 consultations:
    - 04/11.03* - task 1 and task 2
    - 18/25.03* - task 2 and task 3
    - 01/08.04* - task 3

*Date appropriate to the project group

Final submission

- Please prepare zip file (name of the file: Surname1_Surname2_Surname3.zip) including 3 folders: code and report. Please upload your solution using the task assigned in the MS Teams channel.
- **Deadlines: 12.04.2026, 23:59 CEST**
- Presentations: 15.04.2025
    - Group 1 (K.Woźnica) : 14:00
    - Group 2 (W. Grabias): 16:00
    - Group 3 (A. Majczyk): 15:00
    - Group 4 (W. Grabias): 17:00
- If you have any questions, please send us an e-mail: katarzyna.woznica@pw.edu.pl, adam.majczyk@ipipan.waw.pl, wojciech.grabias.stud@pw.edu.pl